

Name:

Roll No:

Mid Semester Examination (Winter 2023)

Branch: **5th Sem, B.Tech. CSE**, Course: **Operating Systems (CSL309)**

Duration: **1.5 hours** Date: **27th September 2023** Max. Marks: **25**

Invigilator
Sign:

***Note:** Write your name & roll no. clearly and the answers in this paper itself. There are total 4 pages. Carefully read the instructions and write your answer in the space provided for that question. For any numerical/solving type question, you can take an extra A4 sheet for any rough work (do not attach it) and after you are sure of the answer, write it in the designated space. If needed, the extra space at the end of paper can also be used to carry over the answer to any question with a note (contd. at end of pg4). In the worst case, if you have to use an extra sheet for writing the answers (along with appropriate note for them), please attach it with the paper but try your best to avoid it as it may incur some penalty of marks deduction.*

Q1. What is the purpose of a trap table in an operating system? What does it contain? When is it consulted? When is it loaded? Answer concisely. [2m]

Q2. Context switching between two threads of execution within the operating system is usually performed by a small assembly language function. In general terms, what does this small function do internally? [2m]

Q3. Consider the following code and answer the questions that follows:

```
int main(void) { pid_t pid = -1;
    for(int i=0; i<4; i++){
        pid = fork();
        if(pid != 0) break;
    }
    waitpid(pid, NULL, 0);
    printf(" %d, %d, %d \n", getpid(), pid, &pid);
    return 0;
}
```

a) Draw the shape (illustration) of process hierarchy that this code produces clearly indicating

different processes and their relationship by appropriate name/IDs. [2m]

- b) What are the values printed by each process? Will they be same or different? [1.5m]
- c) Will the order of printing be consistent or arbitrary? Why? [1m]

Q4. Given a 3-level MLFQ scheduler with a quantum length of 2, 4 and 6 time unit in its highest, middle and lowest queue respectively.

- a) How often would you have to boost jobs back to the highest priority level to guarantee that a single long-running (and potentially starving) job gets at least 20% of the CPU, assuming there are not too many processes competing for CPU? Briefly explain [1.5m]

- b) Lets say 3 jobs arrive together in the order A, B & C at 0 time instant with following workload characteristic.

Process	CPU Burst	I/O Burst	CPU Burst
A	6	4	3
B	4	2	2
C	2	7	1

Assume the priority boost happen every 15 time unit and at that time, the current job is pre-empted and the next scheduled job is run in round-robin order. Illustrate the scheduling in the below table by filling the process name in the corresponding cell of the queue(s) when scheduled for CPU execution at any time instant and leaving other cells blank. [2.5m]

High																										
Mid																										
Low																										
Time->	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Q5. A certain computer system uses segmented paging architecture for memory virtualization. The memory is byte addressable. Both virtual and physical address spaces contain 64K bytes each.

The virtual address space is divided into 8 non-overlapping equal size segments. The memory management unit (MMU) has a hardware segment table, each entry of which contains the physical address of the page table for the segment. Page tables are stored in the main memory and consist of 2 byte page table entries. What is the minimum page size in bytes so that the page table for a segment requires at most one page to store it? Show suitable calculations. [3m]

Q6. a) What does TLB miss handler do when TLB miss occurs during an address translation. Explain your answer based on the valid and present bits of the page table entry. [1.5m]

b) Consider a paging system that uses a one-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 90%, page fault rate is 10%. Assume TLB update time and page table entry update is negligible. What is the average memory access time (in ns) ? [2m]

Q7. Consider a main memory with three page frames and the following sequence of page references: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. Assuming all frames are empty initially, using Least Recently Used (LRU) page replacement policies, populate the below table clearly showing contents of the page frames of memory after each memory reference and when a page fault occurs? [2m]

Reference	3	8	2	3	9	1	6	3	8	9	3	6	2	1	3
Frame 1															
Frame 2															
Frame 3															
Mark P for a fault															

Q8. Consider an OS using a two-level page table. Assume a 32-bit virtual address space, 4 KB pages, page table entry (PTE) is 4 bytes in size. Answer the following questions [2+2 = 4m]

- What is the minimum number of valid virtual pages in an address space such that the multi-level page table becomes its maximum size? Explain.
- What is the minimum number of pages needed for the multi-level page table (including the page directory) when there are 1025 contiguous valid pages somewhere in the virtual address space? Explain.

-----End of Paper-----

Extra Space for carrying over the answer to any question: